Agent Discussion

19 Oct, 2026 · 3:00 PM - 3:30 PM · Ian Pepper

View recap

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Decisions

- Retain **Medical Affairs ownership** of prompts and templates
- Decompose large prompts into **parallel specialised sub-agents**
- Demonstrate a **web application proof of concept** alongside Excel automation

Open questions

- Resolve **production platform** across capability, cost and governance
- Validate **Foundry web research and production readiness**

Agenda

Goal: Review the business problem and demonstrate prompt-engineering solutions for medical asset evaluation.

Meeting notes

Asset Evaluation

- Ian explained that medical affairs teams evaluated potential assets, usually rare or unlicensed drugs without an established trade name, through a question-and-answer process covering medical, regulatory, commercial and competitor considerations. Assets that passed this initial assessment progressed to a second analysis of global disease burden, incidence and prevalence, and the current standard of care
- The asset evaluation process had been organised as a two-stage spreadsheet pipeline: the initial assessment was completed first, and the prevalence analysis was performed only if the asset passed that stage
- Ian explained that the initial asset assessment used a versioned prompt with an Excel template containing the questions to be answered. Claude researched the answers, populated the template, cited the information sources, and covered medical, regulatory, commercial and competitor considerations across multiple worksheets
- The assessment distinguished factual commercial information, such as company and drug names and actual sales, from projections and estimates. These entries were colour-coded using trusted named sources, including S&P Global and Visible Alpha; regular drugs with large patient populations and orphan drugs such as the pre-clinical Angelman syndrome therapy Ian 582 were handled using different rules
- Following a request from Maurice in Medical Affairs, the process added colour coding to indicate whether an asset warranted further review: red entries signalled a problem with the asset, while other colours helped users identify projections and estimates at a glance. A second prompt generated an Excel analysis covering approximately 135 countries, using UN population projections and additional rules to assess global disease-related information
- Ian explained that the global disease-burden analysis used proxy countries when prevalence, incidence or mortality data were sparse or unavailable, particularly in Sub-Saharan Africa and former Soviet republics. Proxies were selected using comparable population characteristics, such as ethnicity; for example, Kazakhstan could use Kyrgyzstan as a proxy. The prompt was configured to specify the appropriate proxy where no disease registry existed, which was especially important

for orphan diseases such as Angelman syndrome and Gaucher syndrome, where fewer than 5,000 patients were typically identified annually

- Ian explained that the team reviewed approximately 50 to 60 assets per year, with the volume reflecting the limited number of new innovative medicines launched rather than a capacity constraint. He said the actual annual figure was probably closer to 50

Prompt Engineering Benefits

- Prompt engineering had reduced the initial asset assessment from one to two days to approximately 15 minutes, and had reduced the prevalence analysis from up to 10 days to approximately 15 minutes, depending on complexity such as the number of competitors
- Ian said the prompt-generated analyses were more accurate than the manual versions because they avoided errors introduced by extensive cutting and pasting between spreadsheets, and they could be rerun easily when the underlying data needed updating
- Ian said Medical Affairs wanted the successful asset-evaluation process converted into an agent, without prescribing a particular agent platform. The proposed solution needed native web search, Excel-file generation and suitable data residency, while allowing Medical Affairs to retain ownership of the prompt and modify its template, questions and content. Copilot Studio and Foundry were being considered, alongside an orchestration-agent concept proposed by Pedro
- In response to Gary's question, Ian explained that the workflow used a large prompt uploaded into Copilot Opus together with the Excel template, after which the model populated the template for the selected asset. Ian said the prompt size might be an issue in Copilot Studio, and explained that the separate KOL-identification workflow had already been divided into four stages that generated intermediate spreadsheets because the required data exceeded Copilot's context window
- Ian confirmed that the prevalence and asset-intelligence prompts were effectively complete and produced accurate, consistent results in Copilot, with each analysis taking approximately 15 minutes. The KOL country-mapping workflow failed because it required searches for 15 KOLs per country and disease area, while the congress-mapping workflow also became too large to complete; Ian expected similar challenges when he began the regulatory-affairs work
- Ian reported that prompt-token usage had fallen from approximately 2,200 tokens to 1,200 during his experiments, although the reduction caused a significant loss of fidelity. He said token counting had become less problematic after cloud models were introduced in Copilot

Disease-Burden Analysis

- Ian said the orphan-disease analysis had initially applied the estimated average of 3.6 patients to every country, which produced an incorrect result. The approach was changed to account for country-specific proxy data, and special handling was also required because one location had four disease registries rather than the expected single registry
- The disease-burden workflow used only publicly available internet data and produced a workbook after users entered the disease area in the prompt. Ownership of the workflow had moved to Medical Affairs
- The prevalence prompt was designed to identify the number of occurrences of a specified condition across the relevant countries. Ian said the standard analysis covered 135 countries, whereas a Perplexity test returned results for 185 countries, including locations such as Vatican City and Tuvalu that were not considered commercially relevant

AI Orchestration

- Ian described a proposed medical-affairs orchestrator that would accept inputs such as an asset evaluation or an RFP, invoke one or more of eight specialised sub-agents according to the request, consolidate their findings into a single data pack for commercial decision-making, and package the results into several audience-specific PowerPoint presentations; Canva was being rolled out for the presentation packaging
- Mark said the spreadsheet-and-prompt approach had outgrown its current method because repeatedly loading large spreadsheets and prompt context would make processing costs increasingly expensive. He recommended moving the underlying information into a database and knowledge graph, supported by background sub-agents that could populate and connect cross-cutting knowledge across companies and countries
- Gary explained that the prompt could be decomposed into dependent and independent components, allowing independent components to run in isolation or in parallel as specialised sub-agents. Mark agreed that Foundry could support the agents and workflows, but cautioned against using one Foundry agent to handle the entire process
- Mark said the existing spreadsheet-and-prompt approach was a sensible way to define and refine the ingress pipeline

before building a large web application, because prompts and workflows could be adapted more easily while the process was still being established

- Mark proposed that the existing templates, prompts and knowledge sources could be placed in a Claude project folder and used to generate a proof-of-concept knowledge graph and web application UI, but no participant committed to carrying out that proposal during the discussion
- Mark proposed creating a deliberately non-functional dashboard prototype in roughly half a day to one day, using existing information, templates and sources to visualise the possible user interface without first building the underlying data structures, agents or workspaces. He said a visual example could help business stakeholders understand the potential value more readily than a verbal explanation and could later support automatically generated proposals and PowerPoint presentations based on templates
- Mark said the scale and value of the opportunity justified investing in a proper web application with a back-end database and knowledge graph, because moving beyond spreadsheets could substantially increase the speed of the analysis and support a first-to-market advantage
- Mark described a potential end-to-end architecture in which workflow ingress and analysis would feed a knowledge graph and a web application, providing a more seamless experience for Medical Affairs as additional Regulatory Affairs use cases emerged
- Gary explained that Claude Code could run scheduled daily tasks to identify and validate assets proactively, but the current setup would run on a user's machine. The longer-term requirement was to productionise the process in an Azure environment so it could operate automatically and return findings when they were identified

Assessment Templates

- Ian showed that the Excel question-and-answer template had evolved into a detailed checklist and summary across medical, regulatory, competitive-landscape, commercial and business-development sections. After generating the workbook, the LLM checked each item to confirm completion and populated details such as the product and brand name, company, regulatory information, competitors, projections, factual data, uses and target population
- The competitive-landscape and commercial/business-development sections used RAG rules to flag areas for review, with amber indicating a potential concern and red indicating that the asset might not be taken forward. For orphan drugs, the template populated additional information such as the target-population projection
- Ian explained that spreadsheets had been used because they reflected how the business had organically developed, with each business unit expected to rely on them, including Regulatory Affairs' so-called EAP database, which was actually a large spreadsheet

Automated Excel Outputs

- Ian confirmed that RAG-style status indicators were applied automatically by the AI to the Excel cells rather than being added manually through cell highlighting. The generated example for Ian 852 showed the template fully populated and marked as completed
- Ian showed that the generated Excel output contained concise, verifiable answers with one or more source URLs. Single URLs were clickable, but Excel did not allow users to click multiple URLs within the same cell
- Ian explained that the completed Excel analysis was first used as a go/no-go gate to determine whether enough people had a particular disease to justify pursuing the asset. Teams then manually extracted relevant sections from the workbook for further work
- Ian explained that the completed analyses fed into commercial, medical and potentially regulatory decks. Teams used the information to make decisions such as how to handle a product, how to store a biologic and where to distribute it; the orchestrator and its component outputs supplied the material used to build these decks
- Ian said the prompts could be rerun to refresh the spreadsheets when updated information was needed. He noted that this was more important for changeable areas than for prevalence data, where disease burden was generally stable unless new registries or more accurate reporting became available

Platform Evaluation

- Mark said Copilot Studio was unlikely to handle the required data volume, based on his tests and its rapid limits, while Gary noted that its changing charging model would make the approach more expensive. Mark also said Microsoft Foundry agents could not directly process Office documents such as Excel and Word files, although Gary clarified that agents could return

structured data for another system to create the outputs

- The teams were currently satisfied with the Excel-based process because it automated the workflow they already used manually. Ian said the immediate priority remained making the next couple of prompts reliable without exceeding the limits encountered when loading large prompts into an LLM tool; a web portal or modern dashboard could be considered as a second phase
- Ian said the current prompts and Excel files were stored on their hard drive. They discussed the need for version control that could support intensive iteration of prompts and templates and could be shared with Medical Affairs, as the existing GIS-based approach worked for Ian but would not work for that team
- Mark recommended moving away from Copilot and Copilot Studio in favour of a Claude Enterprise licence, using Claude Desktop Runner and sub-agents because it would provide greater context-length and orchestration capability. He said the workflow could initially run in a virtual machine, while support for running it on managed devices would be discussed on Wednesday
- Mark suggested hosting the automated workflow in an Azure AI Foundry project so that it would use project-level Foundry capacity rather than an individual user's usage. The participants also noted that the approach depended on whether Foundry supported the required internet access and model capabilities
- Mark said he could not create his own Foundry project or models because he lacked the required permissions, although he had created a small agent in Gary's project. Gary explained that his access came from having a resource group available
- Mark said the interface was difficult to navigate and did not explain why requests silently failed when required model capabilities or settings, such as web search, were unavailable
- Mark said they had been unable to install different prompts into the existing project because they did not know whether they had sufficient project permissions. They noted that a form might be required before the code could be used for further testing
- Mark explained that Anthropic models were restricted to specific Azure regions, so the project might need to run in Sweden Central, where the main AI data centre was located. Gary clarified that this regional requirement meant a new resource had to be created in an appropriate resource group; an API key could then be used to call the model from an application
- Mark said the Playground should still be usable even when the project did not allow an agent to be created, providing a way to test the model separately from agent creation
- Mark found Microsoft Foundry's interface difficult to navigate and said configuring projects, quotas and model capabilities could take longer than building an application. He nevertheless considered Foundry an appropriate back-end platform for running multiple agents and workflows
- Mark noted that the available model's context-window size could affect search quality, particularly when processing large prompts or datasets
- Gary and Mark discussed whether Anthropic models would run on Anthropic infrastructure rather than Microsoft Azure. Mark said the models were licensed in a way that he believed prevented Microsoft or other organisations from running them on their own infrastructure
- Mark said Sweden Central might be the most suitable deployment region because it hosted Microsoft's main AI data centre, while Gary noted that deploying the solution remained difficult
- Gary said the local code-based setup could run on an individual machine and allow participants to see what it was doing, but it did not provide the required enterprise-level capability
- Gary said Microsoft was changing Foundry's charging model to organisation-wide blocks of tokens, and Ian noted that usage was monitored closely. Gary had received a Power Platform alert because Copilot usage had exceeded its allowance, with an apparent reset mechanism that operated only weekly

Filing Discovery

- Gary and Ian discussed extending the process to identify new therapy or disease-area filings automatically. The system could produce a candidate list for users to analyse or ignore, reducing the need to discover opportunities manually through disease-area searches, congresses or Medical Science Liaison contacts

Market Opportunity

- The opportunity focused on high-value, complex therapies such as cell and gene therapies, CAR-T treatments and other biologics rather than generic medicines. Ian cited Kymriah at approximately $500,000 per patient and Zolgensma at approximately $2.5 million per year, highlighting the commercial value of a system that could analyse these therapies

globally

Information Governance

- Mark and Ian considered the combined analysis, templates and prompts commercially sensitive, even though the underlying information came from public sources. They discussed using Teams backed by SharePoint instead of GitHub so access could be restricted to specific people and Medical Affairs could maintain versioned prompts and Excel templates

Workflow Governance

- Ian said they were currently testing the workflow with Pedro, while Ian checked that the templates were populated correctly and that the cited links worked. Pedro was responsible for signing off the medical accuracy; they had already approved the business-development prompt, which Ian said was ready to be made live

Evidence Management

- Ian explained that the workflow extracted relevant text from the cited URLs into spreadsheet cells and included the sources, allowing the analysis to be summarised with supporting evidence. Mark raised the risk that web pages could change over six to twelve months, creating a need to preserve the original point-in-time content rather than relying only on live links
- Ian and Mark discussed that the required model might be better suited to information harvesting than medical reasoning, because the workflow needed to locate and compile relevant evidence about conditions such as Angelman syndrome rather than diagnose or explain them

Web Search Capabilities

- Mark explained that an agent's research depended on the model's training data and that built-in web search often queried an index maintained by OpenAI or Anthropic rather than conducting a live search. As a result, information from the previous 24 hours might be unavailable
- Ian said this limitation had affected the KOL-mapping workflow, which required access to PubMed. Mark explained that a live-search tool such as SearXNG could be connected to ChatGPT or Claude to return current search results as JSON for the agent to use
- Ian explained that Copilot's cloud models used Bing underneath, while direct PubMed access had returned 403 errors because some websites applied restrictions similar to robots.txt rules that blocked AI queries. The use of Bing grounding therefore incurred additional API costs, although the delay in obtaining information was acceptable for disease-prevalence data because those figures did not change frequently
- Ian clarified that the Angelman syndrome workflow did not require a specialist medical model to interpret the disease; it primarily needed to search medical terminology and retrieve comprehensive information from the appropriate sources. Mark noted that models such as MedJammer were specifically designed for medical questions, while Ian cited BioBERT and PubMed-oriented models as other options used in medical-affairs work
- Gary and Mark discussed the difficulty and potential cost of adding the Bing web-search tool to another project, with deployment and access constraints still unresolved
- Mark explained that his self-hosted SearXNG search tool could be configured to return search results as JSON rather than raw web pages, making it suitable for automated workflows that collect and collate information. The approach could also avoid some sites identifying the request as coming from an AI user agent; a Python script or headless browser using Playwright could call the search service and feed the returned results into the model
- Gary and Mark discussed that the current workflow appeared to search first and then retrieve content from individual websites, although some sites returned forbidden errors. Running the search and retrieval locally or in the cloud could make it easier to understand how the process broke down and identify ways to optimise prompts for faster and cheaper execution

Follow-up tasks

	Task	Assigned to	Due date	Bucket
1	Evaluate how to build a web application proof of concept for the asset-analysis workflow by asking Claude what it can produce and reviewing the approach with Medical Affairs (Ian)			
2	Build a proof of concept for the proposed workflow and web application by Wednesday (blocked on: Claude Enterprise licence) (Mark)			
3	Message Tim to arrange Ian's Claude Enterprise licence and help set up access (Mark)			
4	Verify Claude Enterprise licence access by signing in with the Unifi account through Claude Code or the browser (Ian)			
5	Zip the project directory and send it to Mark for POC testing (Ian)			